

PRE-CLASS READING 3 OF 5

The dissenting view, revisited — LeCun in 2026

You already read the paper. LeCun, Y. (2022), "A Path Towards Autonomous Machine Intelligence," OpenReview v0.9.2 — assigned in Class 1 as Pre-Read 5. What is new is what he has done since: AMI Labs, a US\$1.03bn seed, and a sharper public claim. This brief points the old objection at the specific thing you are about to build.

ASSIGNED READING

Do not re-read the paper. The refresher below is enough. Read instead **one or both** of the 2026 items in Source Links — the Brown lecture report and the AMI Labs funding coverage. Both are short. (≈ 12 min)

» source ~12 min • this brief ~10 min

If you did not read it in Class 1, read the 2022 abstract and Section 2 plus the block diagram — about 20 minutes — and then continue here. **Note on currency.** The 2022 paper proposes a research programme, not a working system, and it is a contested position. It is in the Week 4 set as the counter-argument, not as settled fact.

THE REFRESHER — WHAT CLASS 1 ESTABLISHED

- **The efficiency gap.** A teenager learns to drive in ~20 hours; machines need vastly more. LeCun's explanation is that humans carry an internal model of how the world works and machines mostly do not.
- **The proposed fix is an architecture, not a bigger model.** Six modules — configurator, perception, **world model**, cost (fixed intrinsic + trainable critic), actor, short-term memory. The actor proposes candidate actions, the world model simulates their consequences, the cost module scores them, and only then does the system act.
- **JEPA / H-JEPA.** Predict the gist of the future in an abstract representation, not every pixel; stack levels so long-range plans are made coarsely and short-range movements precisely.
- **The disagreement.** Predicting the next token is the wrong kind of prediction — wrong space, and no mechanism for evaluating a plan before committing to it.

Class 1 asked you whether that bet looked right. **This brief asks a narrower and harder question, and it is the only reason the reading is back:** your agent has no world model, no explicit cost, and no planning step. It works anyway. Why?

KEY TAKEAWAY

Every agent in this week's other four readings plans **in tokens** — it writes a sentence about what to do next, and that sentence is produced by predicting likely text. LeCun's architecture names precisely what that leaves out: a predictor of *what happens if I do this*, an explicit objective to optimise against, and a planning step that happens **before** acting rather than one token at a time. His summary judgement at Brown in April 2026 was blunt — current systems "can manipulate language... but in fact, they are completely helpless when it comes to the physical world," and the money being invested on the assumption that language models reach human-level intelligence is "complete BS." You do not have to agree. You do have to be able to say **why the agent you build in class works despite this objection — and where the objection actually bites.**

WHAT IS NEW SINCE CLASS 1

- **He has staked a company on it.** LeCun left Meta in late 2025 and co-founded **AMI Labs** (Advanced Machine Intelligence) in January 2026. It raised **US\$1.03bn** in March 2026 at a US\$3.5bn pre-money valuation — CEO Alexandre LeBrun, backers including Bezos Expeditions, Nvidia, Samsung, Temasek and Toyota Ventures. First disclosed partner: Nabla, in digital health.
- **The claim got sharper and the tone got harder.** In 2022 this was a research proposal. In 2026 it is a public bet with a valuation attached and a named competitor set — which is worth noticing on its own, because it is the same move the other three readings describe from the other side.
- **The evidence moved too — in the other direction.** Between Class 1 and now you have read what agents actually did in 2024–26. The strong form of the 2022 claim (token-predictors cannot plan or reason) has aged badly. The weak form — that they plan *inefficiently* and *unreliably*, and that the reliability ceiling is structural — has aged rather well. Distinguishing the two is the whole exercise.

WHERE THE OBJECTION BITES — AND WHERE IT DOES NOT

Our reading — this test is ours, not LeCun's. Ask three things of the task your agent is doing: are the actions **symbolic** (an API call, a query, a message — not a physical movement)? Are they **fast** (milliseconds to seconds, so trying is cheaper than simulating)? Are they **reversible** (a wrong search costs tokens, not a broken machine)? Where all three hold, you do not need a world model, because **the environment is the world model** — you do not simulate

the consequence of an action, you execute it and read the observation. That is exactly what the ReAct loop does, and it is why it works.

Where any of the three fails — physical actuation, expensive or slow actions, irreversible ones (a payment, a production deploy, a clinical order) — you are asking a system with no simulator to plan by trial in the real world, and the objection lands with full force. Notice that this is the same line that separates the agent tasks industry has shipped from the ones it has not.

And one place it bites inside software. Your agent has no representation of the run to update — its only state is the transcript. That is why context fills, why every turn re-reads the whole history from the beginning, and why long runs have to be **compacted**: a summary written by the same next-token predictor whose adequacy is under dispute, deciding at turn 12 what will still matter at turn 30. Whatever it drops is gone. **Our reading** — this is the objection arriving as an engineering problem rather than a philosophical one, and it is why reliability degrades with run length even on symbolic, reversible tasks.

Feeds Class 4: C1 (what a control loop leaves out) • C2 (loops and failed plans as consequences, not bugs) • C3 (the strategic bet — is thinking-out-loud-in-text an efficient way to plan, or only the one we have?).

READING FOCUS QUESTIONS — bring your answers to class

Three questions, and none of them is Class 1's question. There are no wrong answers — these surface how you think. A clue sits under each; fuller reasoning follows.

- Your Capsule 2 agent will fail in at least two ways: it will repeat the same action, and it will act confidently on a bad tool result. One of those is the absence of a world model. Which — and what is the other one the absence of?

Clue. Ask what a simulator would have told the agent in each case. For one failure it would have said “you already tried this and nothing changed” — that is a missing model of state. For the other it would have said nothing useful at all, because the problem is that the observation was wrong, not that the consequence was unforeseen. Naming the second one correctly is the point of the question.

- If the environment can serve as the world model — execute and observe instead of simulating — what exactly are you paying for that substitution, and in what currency?

Clue. List what a simulation gives you for free that execution charges for: time, money, side effects, and the ability to try a plan you would never dare run. Then price each in a domain you know. The answer is not “nothing” — it is a specific bill, and one of the four items on it is the one that stops this substitution working outside software.

- Investors put US\$1.03bn behind the claim that this whole approach is a dead end, in the same year enterprises put far more behind deploying it. Both cannot be straightforwardly right. What would have to be true for both to be reasonable bets?

Clue. Resist picking a winner. Look for the assumption each side could hold without contradicting the other — most likely something about time horizon and what counts as the goal. Then ask which bet you are personally making by choosing what to learn this year, because you are making one.

HOW TO THINK ABOUT THESE — read after you have had a go

Not model answers — the line of reasoning we will pick up in class.

Q1 — One is a missing world model; the other is not

Repetition is the missing world model. An agent that reissues the same search has no representation of whether its last action changed anything; it cannot ask “what did that do?” because there is nothing in the architecture that predicts or tracks state. That is LeCun's point, arriving as a bug in your notebook. **Acting on a bad tool result is a different failure** — a grounding and evaluation failure. The agent's picture of the world is wrong because the observation was wrong; a world model would have faithfully simulated forward from the same bad premise. This distinction matters practically: the first is fixed in the loop (dedupe, step cap, state tracking), the second only by improving what the tools return and checking output against sources. **Our reading** — treating both as “the agent is dumb” is how teams ship the wrong fix.

Q2 — What execution charges that simulation gives away

Four things. **Time** — every hypothesis costs a round trip, so an agent can explore three options where a simulator could score three hundred. **Money** — each trial is billed tokens and API calls, and Reading 2's quadratic makes the later trials the expensive ones. **Side effects** — a simulated action is free to be wrong; an executed one may have sent an email, charged a card or written a row. **Counterfactuals you would never run** — you cannot execute the plan that might destroy something in order to find out that it would. The first two are inconveniences and get cheaper every year. The third and fourth are the reason this substitution works beautifully for read-only software tasks and stops working the moment an agent touches money, hardware or a patient. **Our reading** — that boundary, not the model's intelligence, is what actually gates enterprise deployment today.

Q3 — Both bets can be reasonable

They are bets on different horizons about different goals. The enterprise bet is: token-native agents are useful **now**, on symbolic and reversible tasks, and useful is a low bar that pays this year. LeCun's bet is: the path to systems that plan robustly in the physical world does not run through here, and that destination is worth a decade. Both can hold simultaneously, because “commercially valuable in 2026” and “the route to human-level machine intelligence” are unrelated tests, and the field routinely confuses them. **Our reading** — the honest position for a practitioner is dual: build with what works, and keep the objection live as your explanation for why reliability plateaus where it does, rather than assuming the next model release dissolves it. What you should not do is hold the strong form of either claim, because the evidence of the last two years has embarrassed both.

APPENDIX — THE TERMS THIS BRIEF TURNS ON

The rest of the architecture — the six modules, perception, the cost module, the actor, JEPA and H-JEPA — is defined in your Class 1 Pre-Read 5 brief. Go back to that one page rather than re-reading it here.

World model

- **Plain.** An internal simulator: given where things stand and what I might do, what happens next?
Technically. Defined here rather than by cross-reference, because the whole brief turns on its absence. A learned module that predicts future world state from current state plus a **proposed** action, and fills in what perception cannot observe. Its function is to make an action evaluable *without executing it*.

Objective-driven AI

- **Plain.** Decide what you want, imagine several ways to get it, score them, then act.
Technically. Inference-time optimisation of an action **sequence** against an explicit cost, using the world model to score candidates before any of them runs. Contrast the ReAct loop, which picks one action at a time and discovers its cost by executing it. The difference is not intelligence; it is *where the evaluation happens* — inside the system, or out in the world.

AMI Labs

- **Plain.** LeCun's company, founded to build world models instead of language models.
Technically. Advanced Machine Intelligence; founded Jan 2026; US\$1.03bn seed announced 9 Mar 2026 at US\$3.5bn pre-money; CEO Alexandre LeBrun; investors include Bezos Expeditions, Nvidia, Samsung, Temasek, Toyota Ventures. First disclosed partner: Nabla (digital health).

SOURCE LINKS

Brown University — lecture report, 1 April 2026 · www.brown.edu/news/2026-04-01/yann-lecun-artificial-intelligence-pioneer (assigned)

TechCrunch — AMI Labs raises US\$1.03bn, 9 March 2026 · techcrunch.com/2026/03/09/yann-lecuns-ami-labs-raises-1-03-billion-to-build-world-models/ (assigned)

Paper — “A Path Towards Autonomous Machine Intelligence” (OpenReview) · openreview.net/forum?id=BZ5a1r-kVsf (2022; read in Class 1 — reference only)

MIT Technology Review — LeCun's contrarian bet, 22 January 2026 · www.technologyreview.com/2026/01/22/1131661/yann-lecuns-new-venture-ami-labs/ (optional; may be paywalled)

Companion · **Class 1 Pre-Read 5** is the full treatment of the paper and the source for the architecture vocabulary; **Reading 2** (ReAct) is the thing being criticised; **Reading 4** (Hassabis) is a third position — build the agents now, and let world models arrive inside them.

CAVEATS

1. Provenance. Quoted phrases, the 2022 architecture and the 2026 funding facts are from the sources named above. Anything marked **Our reading** is this brief's interpretation — the “symbolic, fast, reversible” test, the “environment as world model” framing and the two-bets reading are ours, not LeCun's.
2. This is a contested position, included as the counter-argument in the set. Presenting it does not endorse it — and the strongest agent results of 2025–26 are evidence against the strong form of the claim.
3. The 2026 material is reported from press coverage (Brown University, TechCrunch, MIT Technology Review), not from primary company filings. Quotes are as reported. Funding facts were correct as at August 2026.
4. This summary was generated and fact-checked using Anthropic models. Flag anything that looks wrong.